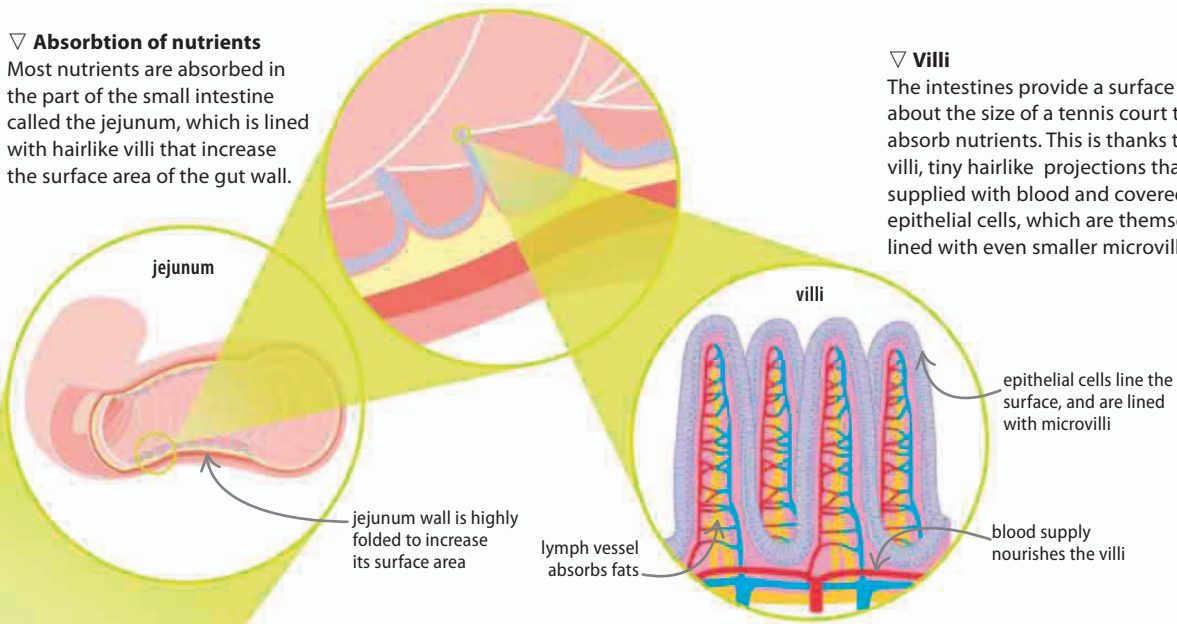


▽ Absorption of nutrients

Most nutrients are absorbed in the part of the small intestine called the jejunum, which is lined with hairlike villi that increase the surface area of the gut wall.

▽ Villi

The intestines provide a surface area about the size of a tennis court to absorb nutrients. This is thanks to the villi, tiny hairlike projections that are supplied with blood and covered in epithelial cells, which are themselves lined with even smaller microvilli.



Digestive chemicals

Digestion is both a physical and a chemical process. It starts in the mouth, when the teeth mechanically grind up the food. This pulp is mixed with saliva, which contains enzymes that work on the food. Enzymes target specific foods, dividing complex foods, such as starches and proteins, into smaller, simpler ingredients—sugar and amino acids respectively—that can be absorbed more easily.

▽ Chemical chart

A range of digestive chemicals work on the food at each stage of its journey through the gut. The chemicals each have a specific role to play in breaking down the food, and are produced by glands and organs along the alimentary canal.

	Enzyme or other chemical	Function	Produced by
Mouth	lipase (enzyme)	digests fats	salivary gland
	amylase (enzyme)	digests starch	salivary gland
	mucin	lubricates food	salivary gland and gut lining
	bicarbonate (enzyme)	kills bacteria, neutralizes acids	salivary gland
Stomach	pepsin (enzyme)	digests proteins	stomach cells
	hydrochloric acid	kills bacteria	stomach cells
	rennin (enzyme)	digests milk	stomach cells
Small intestine	bile	aids digestion of fats	liver, via gall bladder
	trypsin (enzyme)	digest proteins	pancreas
	nuclease (enzyme)	digest nucleic acids	pancreas
	phospholipase (enzyme)	digests fats	pancreas
	amylase (enzyme)	digests starches	pancreas
	sucrase (enzyme)	digests sucrose	duodenum
	lactase (enzyme)	digests lactose (sugar found in milk)	duodenum
	maltase (enzyme)	digests maltose (sugar found in starch)	duodenum